(An Autonomous Institution, Affiliated to Anna University, Chennai)

Department of Artificial Intelligence and Data Science

V Semester B. Tech AIDS

AD23532 – Principles of Data Science

Academic Year: 2026-2027 Batch: 2024-2028

Name: muthukutti R | Register No: 241801175 | Date: 10/6/26

Ex 4: Information Retrieval using NLTK

Learning Objective:

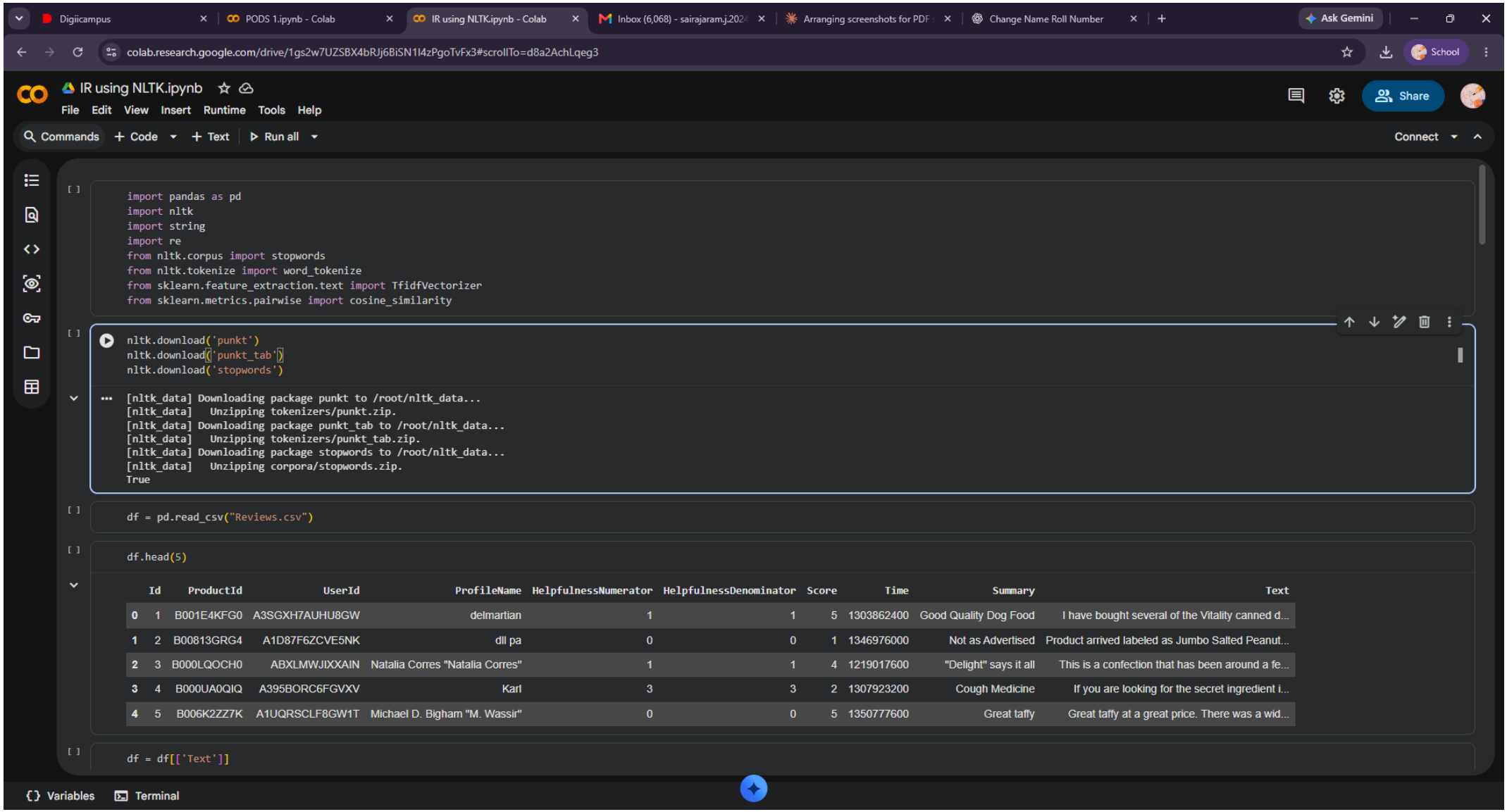
The objective of this experiment is to enable students to understand the text preprocessing pipeline and implement a basic information retrieval system using TF-IDF vectorization, Cosine Similarity, NLTK.

Problem Statement:

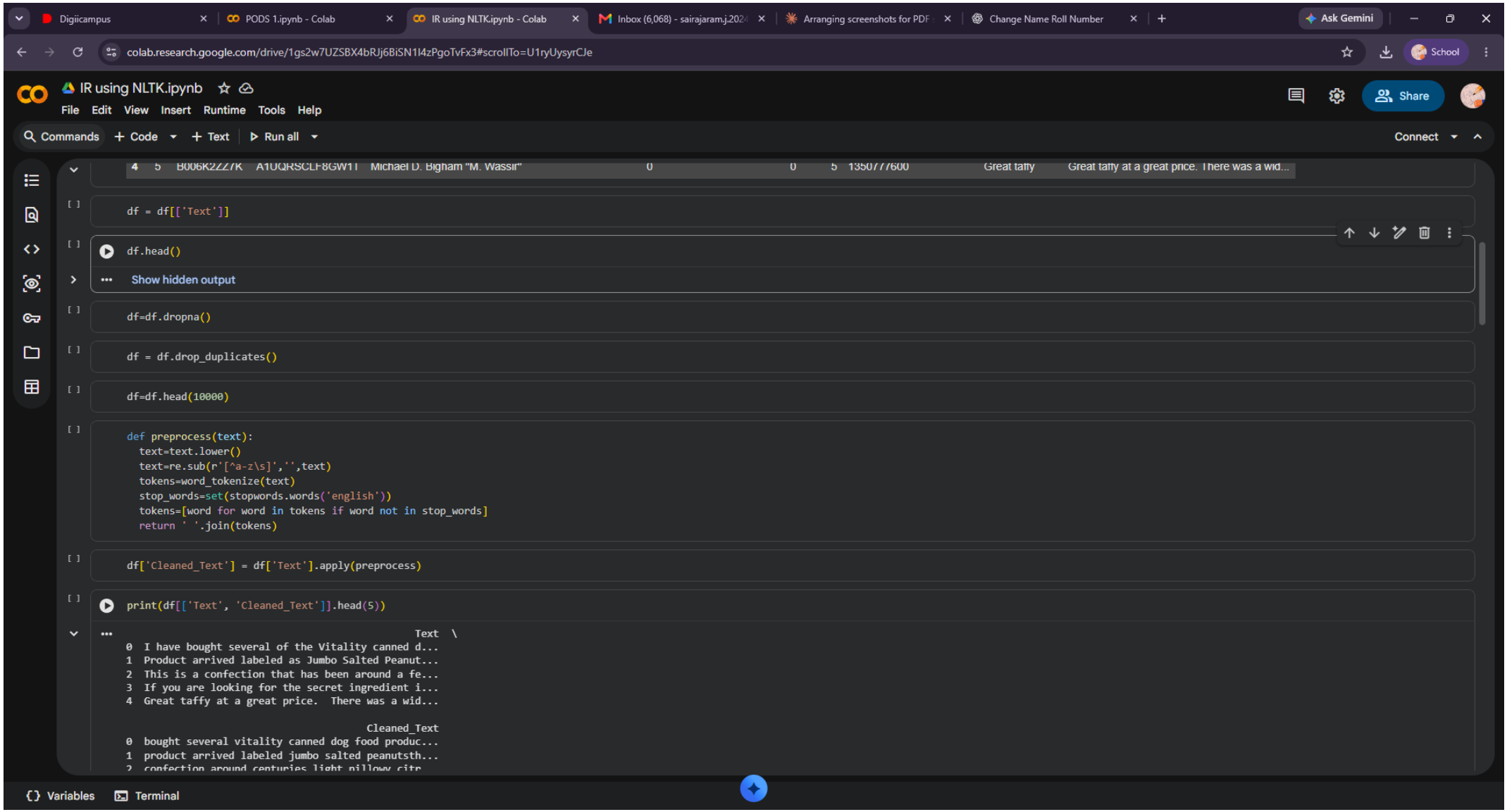
An e-commerce company receives thousands of customer reviews for its products every day. Customer support executives often need to retrieve reviews that are similar to a user's search query to quickly understand customer feedback and address common concerns.

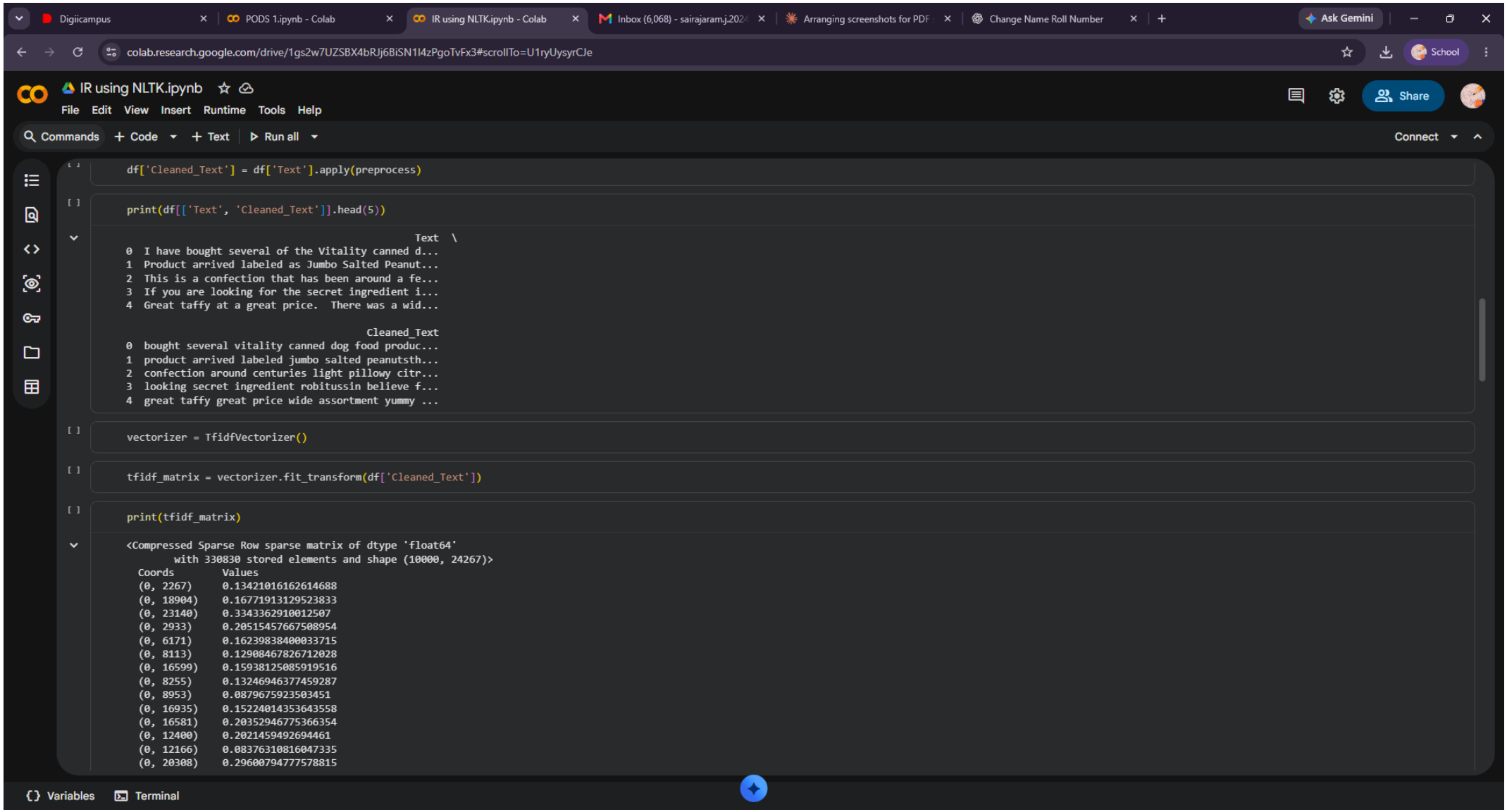
Implementation Evidence:

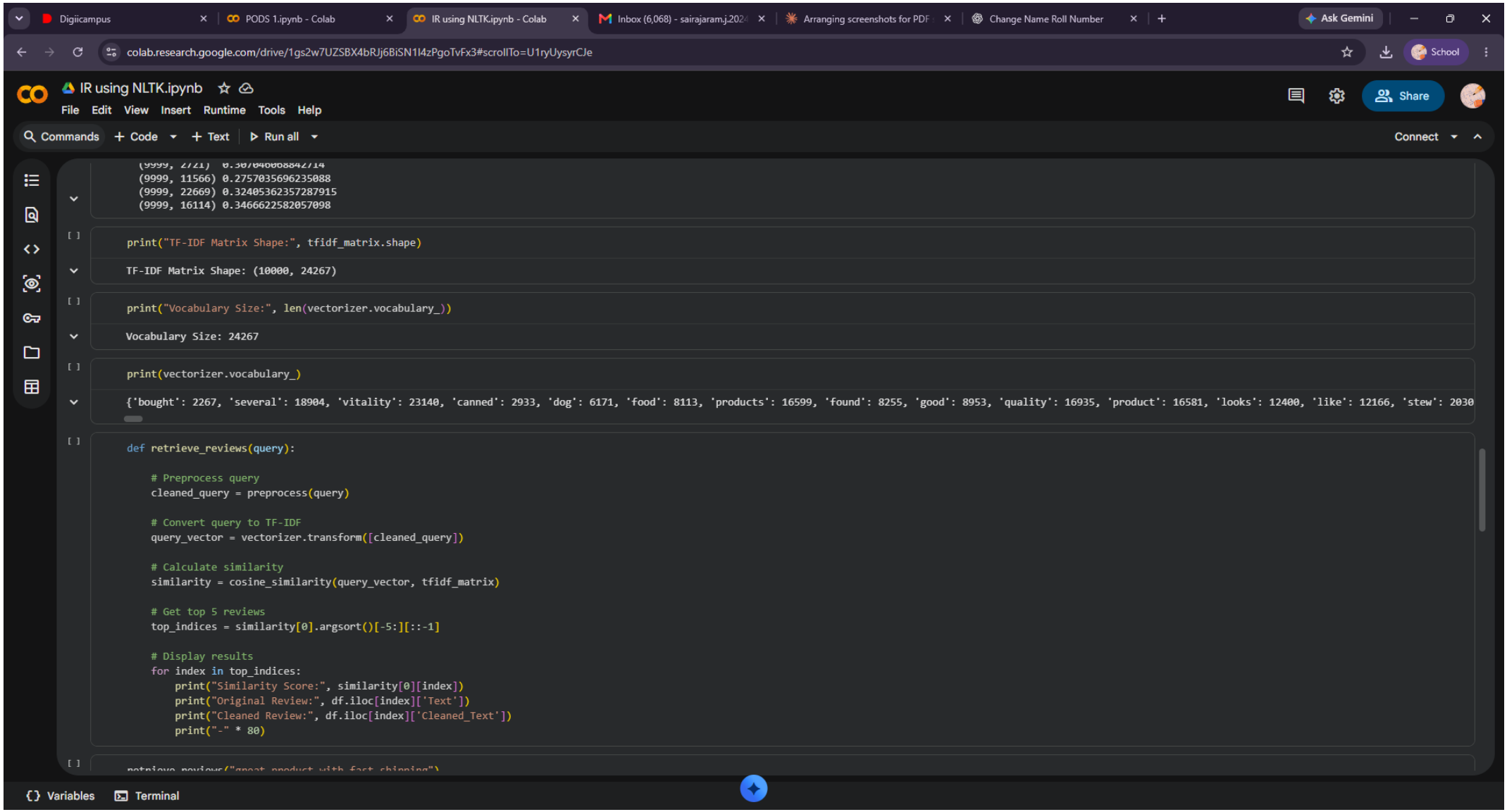
The following pages contain the submitted Google Colab screenshots showing the code and outputs.

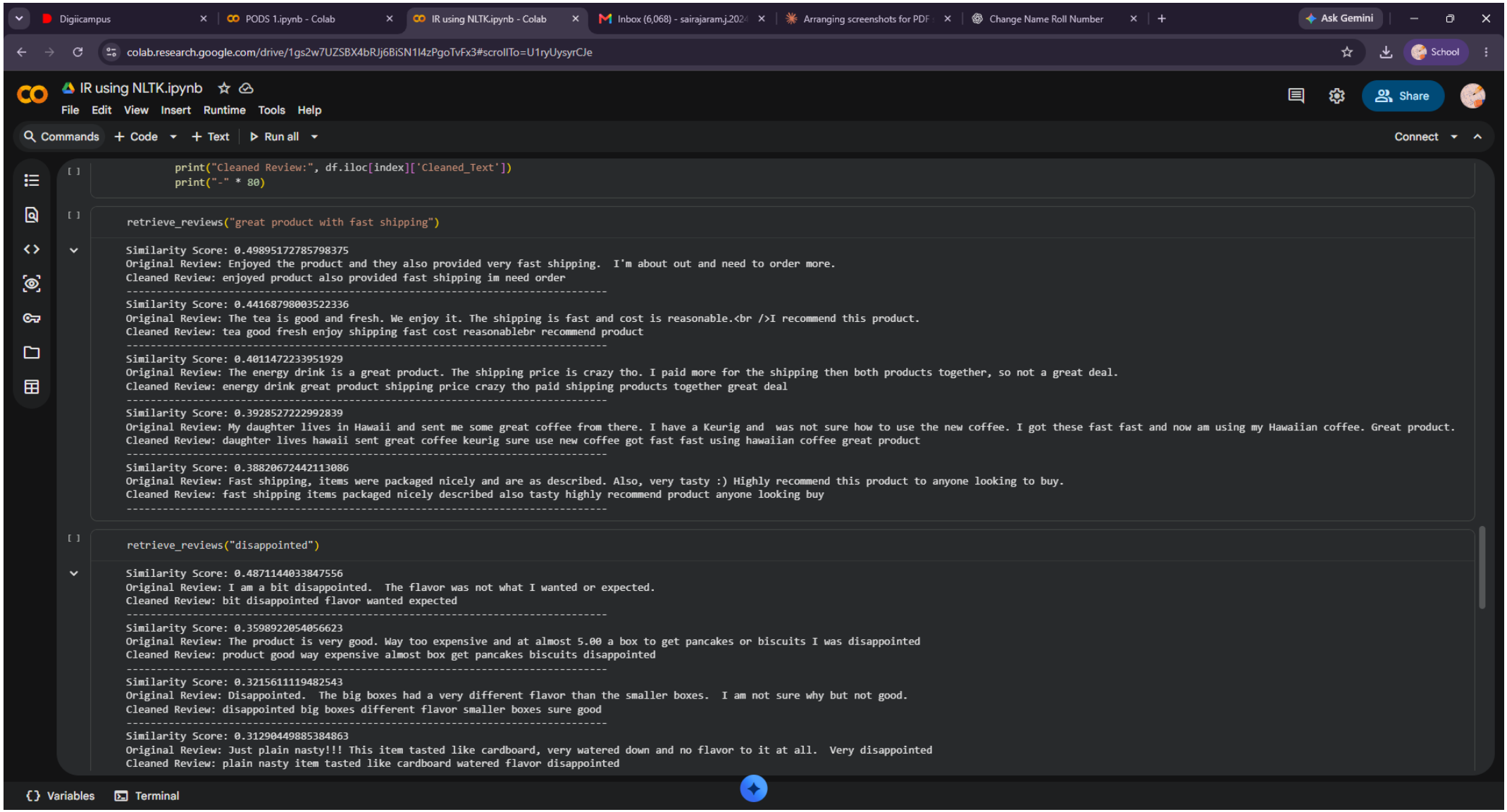


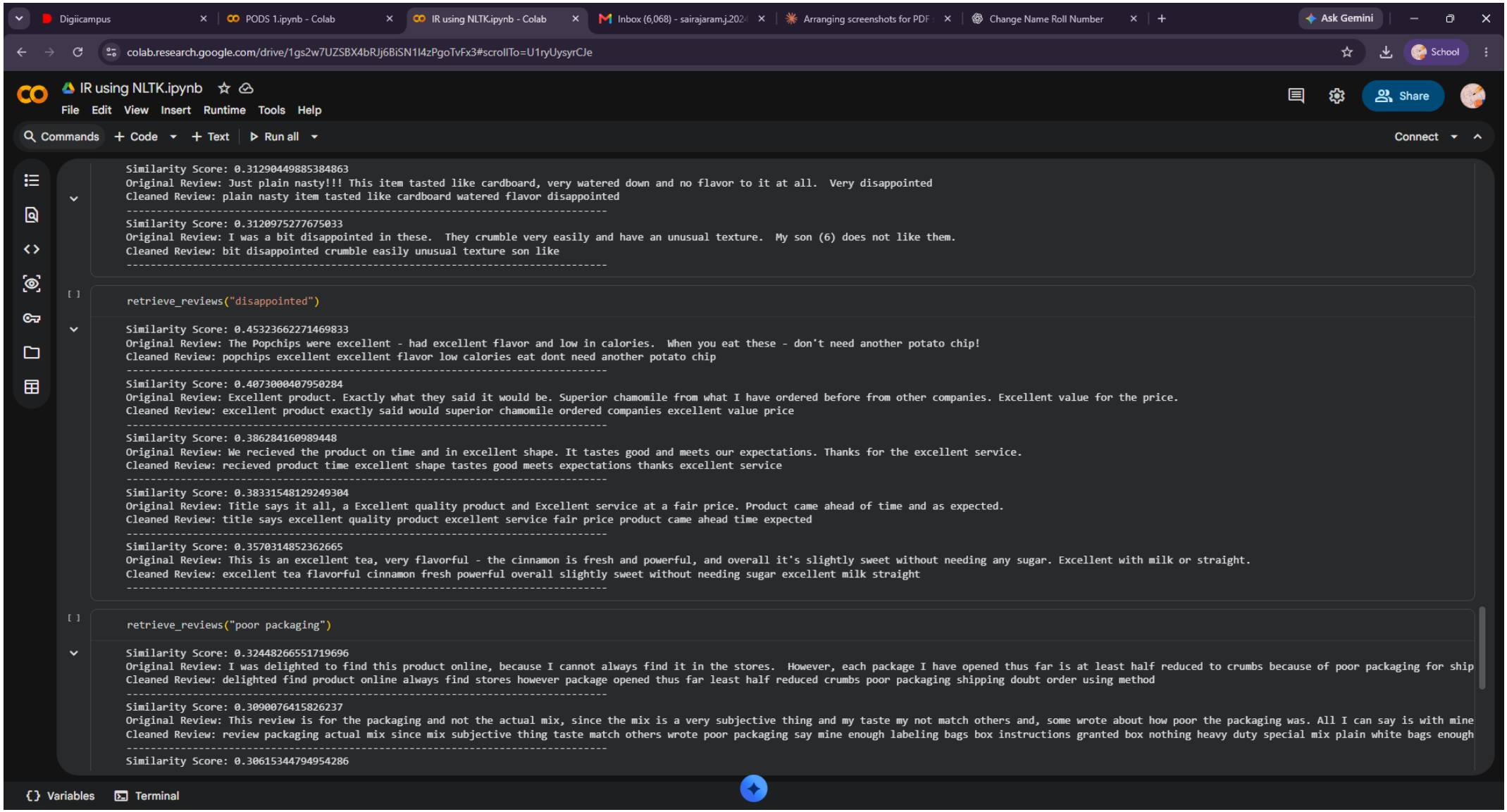
	Id	ProductId	UserId	ProfileName	HelpfulnessNumerator	HelpfulnessDenominator	Score	Time	Summary	Text
0	1	B001E4KFG0	A3SGXH7AUHU8GW	delmartian	1	1	5	1303862400	Good Quality Dog Food	I have bought several of the Vitality canned d...
1	2	B00813GRG4	A1D87F6ZCVE5NK	dll pa	0	0	1	1346976000	Not as Advertised	Product arrived labeled as Jumbo Salted Peanut...
2	3	B000LQOCH0	ABXLMWJIXXAIN	Natalia Corres "Natalia Corres"	1	1	4	1219017600	"Delight" says it all	This is a confection that has been around a fe...
3	4	B000UA0QIQ	A395BORC6FGVXV	Karl	3	3	2	1307923200	Cough Medicine	If you are looking for the secret ingredient i...
4	5	B006K2ZZ7K	A1UQRSCLF8GW1T	Michael D. Bigham "M. Wassir"	0	0	5	1350777600	Great taffy	Great taffy at a great price. There was a wid...

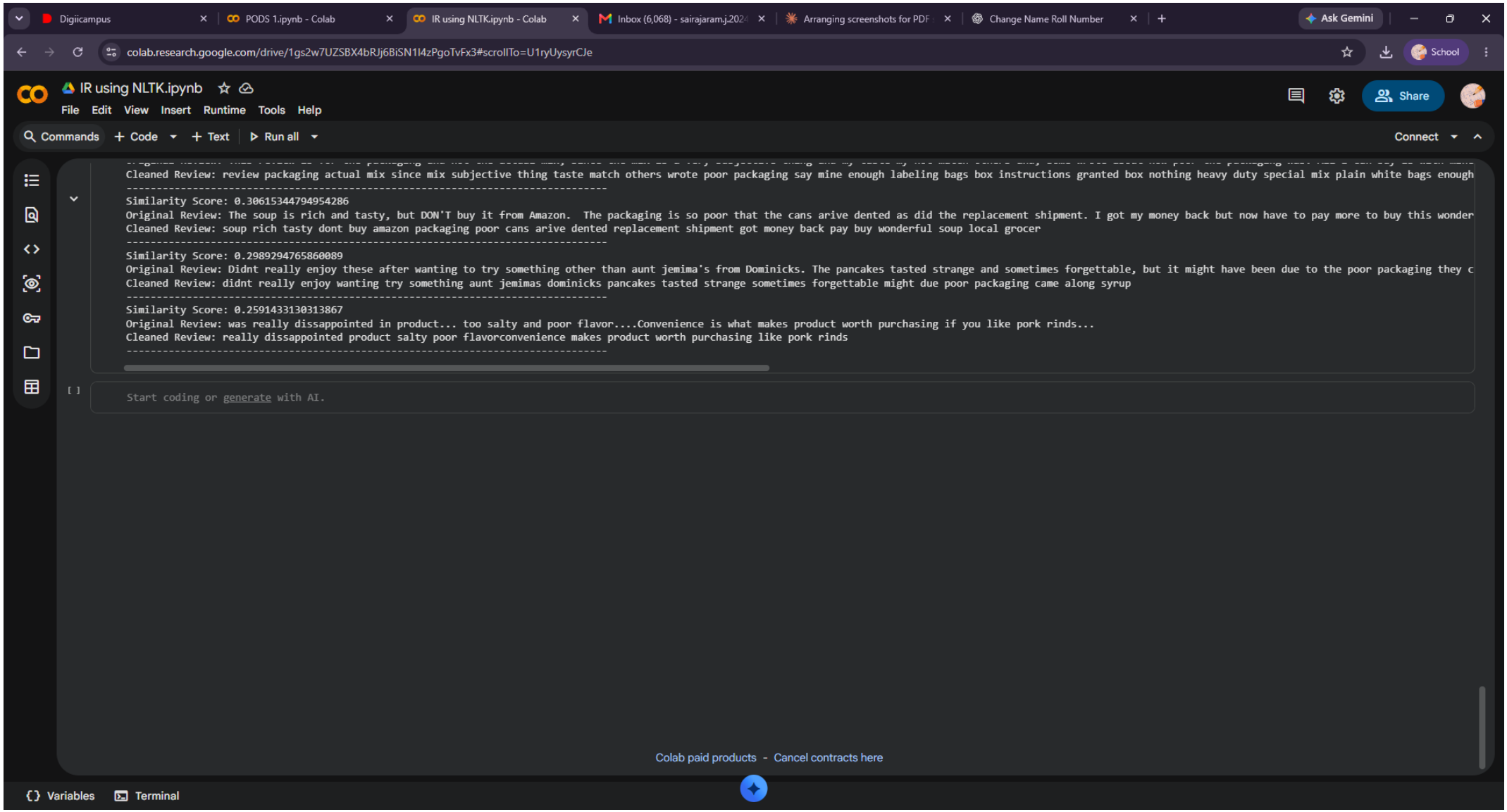

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4 5 B006K2ZZ7K A1UQRSCLF8GW1T Michael D. Bigham "M. Wassir" 0 0 5 1350777600 Great taffy Great taffy at a great price. There was a wid...
df = df[['Text']]
df.head()
Show hidden output
df=df.dropna()
df = df.drop_duplicates()
df=df.head(10000)
def preprocess(text):
text=text.lower()
text=re.sub(r'[^a-z\s]','',text)
tokens=word_tokenize(text)
stop_words=set(stopwords.words('english'))
tokens=[word for word in tokens if word not in stop_words]
return ' '.join(tokens)
df['Cleaned_Text'] = df['Text'].apply(preprocess)
print(df[['Text', 'Cleaned_Text']].head(5))
Text \
0 I have bought several of the Vitality canned d...
1 Product arrived labeled as Jumbo Salted Peanut...
2 This is a confection that has been around a fe...
3 If you are looking for the secret ingredient i...
4 Great taffy at a great price. There was a wid...
Cleaned_Text
0 bought several vitality canned dog food produc...
1 product arrived labeled jumbo salted peanutsth...
Variables
Terminal


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File Edit View Insert Runtime Tools Help
Share
Commands + Code + Text Run all
Connect
df['Cleaned_Text'] = df['Text'].apply(preprocess)
print(df[['Text', 'Cleaned_Text']].head(5))
Text \
0 I have bought several of the Vitality canned d...
1 Product arrived labeled as Jumbo Salted Peanut...
2 This is a confection that has been around a fe...
3 If you are looking for the secret ingredient i...
4 Great taffy at a great price. There was a wid...
Cleaned_Text
0 bought several vitality canned dog food produc...
1 product arrived labeled jumbo salted peanutsth...
2 confection around centuries light pillowy citr...
3 looking secret ingredient robitussin believe f...
4 great taffy great price wide assortment yummy ...
vectorizer = TfidfVectorizer()
tfidf_matrix = vectorizer.fit_transform(df['Cleaned_Text'])
print(tfidf_matrix)
<Compressed Sparse Row sparse matrix of dtype 'float64'
with 330830 stored elements and shape (10000, 24267)>
Coords Values
(0, 2267) 0.13421016162614688
(0, 18904) 0.16771913129523833
(0, 23140) 0.3343362910012507
(0, 2933) 0.20515457667508954
(0, 6171) 0.16239838400033715
(0, 8113) 0.12908467826712028
(0, 16599) 0.15938125085919516
(0, 8255) 0.13246946377459287
(0, 8953) 0.0879675923503451
(0, 16935) 0.15224014353643558
(0, 16581) 0.20352946775366354
(0, 12400) 0.2021459492694461
(0, 12166) 0.08376310816047335
(0, 20308) 0.29600794777578815
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Share
Commands + Code + Text Run all
Connect
(9999, 11566) 0.2757035696235088
(9999, 22669) 0.32405362357287915
(9999, 16114) 0.3466622582057098
print("TF-IDF Matrix Shape:", tfidf_matrix.shape)
TF-IDF Matrix Shape: (10000, 24267)
print("Vocabulary Size:", len(vectorizer.vocabulary_))
Vocabulary Size: 24267
print(vectorizer.vocabulary_)
{'bought': 2267, 'several': 18904, 'vitality': 23140, 'canned': 2933, 'dog': 6171, 'food': 8113, 'products': 16599, 'found': 8255, 'good': 8953, 'quality': 16935, 'product': 16581, 'looks': 12400, 'like': 12166, 'stew': 2030
def retrieve_reviews(query):
Preprocess query
cleaned_query = preprocess(query)
Convert query to TF-IDF
query_vector = vectorizer.transform([cleaned_query])
Calculate similarity
similarity = cosine_similarity(query_vector, tfidf_matrix)
Get top 5 reviews
top_indices = similarity[0].argsort()[-5:][::-1]
Display results
for index in top_indices:
print("Similarity Score:", similarity[0][index])
print("Original Review:", df.iloc[index]['Text'])
print("Cleaned Review:", df.iloc[index]['Cleaned_Text'])
print("-" * 80)
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Connect
print("Cleaned Review:", df.iloc[index]['Cleaned_Text'])
print("-" * 80)
retrieve_reviews("great product with fast shipping")
Similarity Score: 0.49895172785798375
Original Review: Enjoyed the product and they also provided very fast shipping. I'm about out and need to order more.
Cleaned Review: enjoyed product also provided fast shipping im need order
Similarity Score: 0.44168798003522336
Original Review: The tea is good and fresh. We enjoy it. The shipping is fast and cost is reasonable.
I recommend this product.
Cleaned Review: tea good fresh enjoy shipping fast cost reasonablebr recommend product
Similarity Score: 0.4011472233951929
Original Review: The energy drink is a great product. The shipping price is crazy tho. I paid more for the shipping then both products together, so not a great deal.
Cleaned Review: energy drink great product shipping price crazy tho paid shipping products together great deal
Similarity Score: 0.3928527222992839
Original Review: My daughter lives in Hawaii and sent me some great coffee from there. I have a Keurig and was not sure how to use the new coffee. I got these fast fast and now am using my Hawaiian coffee. Great product.
Cleaned Review: daughter lives hawaii sent great coffee keurig sure use new coffee got fast fast using hawaiian coffee great product
Similarity Score: 0.38820672442113086
Original Review: Fast shipping, items were packaged nicely and are as described. Also, very tasty :) Highly recommend this product to anyone looking to buy.
Cleaned Review: fast shipping items packaged nicely described also tasty highly recommend product anyone looking buy
retrieve_reviews("disappointed")
Similarity Score: 0.4871144033847556
Original Review: I am a bit disappointed. The flavor was not what I wanted or expected.
Cleaned Review: bit disappointed flavor wanted expected
Similarity Score: 0.3598922054056623
Original Review: The product is very good. Way too expensive and at almost 5.00 a box to get pancakes or biscuits I was disappointed
Cleaned Review: product good way expensive almost box get pancakes biscuits disappointed
Similarity Score: 0.3215611119482543
Original Review: Disappointed. The big boxes had a very different flavor than the smaller boxes. I am not sure why but not good.
Cleaned Review: disappointed big boxes different flavor smaller boxes sure good
Similarity Score: 0.31290449885384863
Original Review: Just plain nasty!!! This item tasted like cardboard, very watered down and no flavor to it at all. Very disappointed
Cleaned Review: plain nasty item tasted like cardboard watered flavor disappointed
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Share
Commands + Code + Text Run all
Connect
Similarity Score: 0.31290449885384863
Original Review: Just plain nasty!!! This item tasted like cardboard, very watered down and no flavor to it at all. Very disappointed
Cleaned Review: plain nasty item tasted like cardboard watered flavor disappointed
Similarity Score: 0.3120975277675033
Original Review: I was a bit disappointed in these. They crumble very easily and have an unusual texture. My son (6) does not like them.
Cleaned Review: bit disappointed crumble easily unusual texture son like
retrieve_reviews("disappointed")
Similarity Score: 0.45323662271469833
Original Review: The Popchips were excellent - had excellent flavor and low in calories. When you eat these - don't need another potato chip!
Cleaned Review: popchips excellent excellent flavor low calories eat dont need another potato chip
Similarity Score: 0.4073000407950284
Original Review: Excellent product. Exactly what they said it would be. Superior chamomile from what I have ordered before from other companies. Excellent value for the price.
Cleaned Review: excellent product exactly said would superior chamomile ordered companies excellent value price
Similarity Score: 0.386284160989448
Original Review: We recieved the product on time and in excellent shape. It tastes good and meets our expectations. Thanks for the excellent service.
Cleaned Review: recieved product time excellent shape tastes good meets expectations thanks excellent service
Similarity Score: 0.38331548129249304
Original Review: Title says it all, a Excellent quality product and Excellent service at a fair price. Product came ahead of time and as expected.
Cleaned Review: title says excellent quality product excellent service fair price product came ahead time expected
Similarity Score: 0.3570314852362665
Original Review: This is an excellent tea, very flavorful - the cinnamon is fresh and powerful, and overall it's slightly sweet without needing any sugar. Excellent with milk or straight.
Cleaned Review: excellent tea flavorful cinnamon fresh powerful overall slightly sweet without needing sugar excellent milk straight
retrieve_reviews("poor packaging")
Similarity Score: 0.32448266551719696
Original Review: I was delighted to find this product online, because I cannot always find it in the stores. However, each package I have opened thus far is at least half reduced to crumbs because of poor packaging for ship
Cleaned Review: delighted find product online always find stores however package opened thus far least half reduced crumbs poor packaging shipping doubt order using method
Similarity Score: 0.3090076415826237
Original Review: This review is for the packaging and not the actual mix, since the mix is a very subjective thing and my taste my not match others and, some wrote about how poor the packaging was. All I can say is with mine
Cleaned Review: review packaging actual mix since mix subjective thing taste match others wrote poor packaging say mine enough labeling bags box instructions granted box nothing heavy duty special mix plain white bags enough
Similarity Score: 0.30615344794954286
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Cleaned Review: review packaging actual mix since mix subjective thing taste match others wrote poor packaging say mine enough labeling bags box instructions granted box nothing heavy duty special mix plain white bags enough
Similarity Score: 0.30615344794954286
Original Review: The soup is rich and tasty, but DON'T buy it from Amazon. The packaging is so poor that the cans arive dented as did the replacement shipment. I got my money back but now have to pay more to buy this wonder
Cleaned Review: soup rich tasty dont buy amazon packaging poor cans arive dented replacement shipment got money back pay buy wonderful soup local grocer
Similarity Score: 0.2989294765860089
Original Review: Didnt really enjoy these after wanting to try something other than aunt jemima's from Dominicks. The pancakes tasted strange and sometimes forgettable, but it might have been due to the poor packaging they c
Cleaned Review: didnt really enjoy wanting try something aunt jemimas dominicks pancakes tasted strange sometimes forgettable might due poor packaging came along syrup
Similarity Score: 0.2591433130313867
Original Review: was really dissappointed in product... too salty and poor flavor....Convenience is what makes product worth purchasing if you like pork rinds...
Cleaned Review: really dissappointed product salty poor flavorconvenience makes product worth purchasing like pork rinds
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